

SUVRADIP GHOSH

West Bengal, India | suvrawork03@gmail.com | +91 8653056408 | linkedin.com/in/suvradip01 | github.com/Suvradip01

SUMMARY

Software Developer working across full-stack development and machine learning, with experience in building production software and applying modern AI techniques to real-world tasks, from experimentation to deployment and maintenance.

SKILLS

Programming Languages: Python, Java, JavaScript, C
Frontend: React.js, HTML, CSS, Tailwind CSS
Backend: Node.js, Express.js, FastAPI, REST APIs
Databases: MongoDB Atlas, SQL
Tools & Platforms: Git, Docker, Clerk, Convex, ImageKit
AI/ML: PyTorch, TensorFlow, Scikit-Learn, OpenCV, SHAP

PROJECTS

Insight-ATS: AI-Assisted Hiring and Project-Based Evaluation Platform Live | [GitHub](#)

Tech Stack: React.js, Recharts, FastAPI, PyTorch, BERT, RoBERTa, DistilBERT, SHAP

- Built a full-stack AI candidate-evaluation ATS platform with a FastAPI backend integrating **3 transformer models** for entity extraction, Resume-JD matching, extended project complexity scoring, and SHAP explainability.
- Fine-tuned and evaluated 3 models: **BERT-NER** entity extractor (1000 resumes, **0.75 F1**), **RoBERTa** resume-JD semantic matcher (8000 resume-JD pairs, **0.78 Macro F1**), and **DistilBERT** project-complexity evaluator (**0.93 Macro F1**) using 3×5 repeated K-Fold validation.
- Optimized multi-resume inference latency by **51.1% (6.62s to 3.24s)** by designing a FastAPI pipeline with lazy loaded singleton orchestration and in-memory cached model instances serving **4-dimensional** ATS scores.
- Reduced client-to-backend API calls by **40%** via shared-JD batch screening and deployed the system as a production full-stack application with Clerk authentication on Hugging Face Spaces.

CodeNest: AI-Powered Cloud IDE and Review Platform Live | [GitHub](#)

Tech Stack: React.js, Express.js, MongoDB Atlas, Docker, Groq Llama-3.3-70B

- Built a microservice-based cloud IDE supporting **20+ concurrent users**, executing JavaScript, Python, Java, and C in isolated Docker containers with resource quotas, network restrictions, and **5-second** execution timeouts.
- Engineered a queue-based execution system with a **25-job buffer** and **2 worker processes** to prevent resource exhaustion and ensure reliable code execution under concurrent workloads.
- Integrated **Groq-hosted Llama-3.3-70B** for AI-assisted code reviews, debugging, Mermaid flowchart generation, and code analysis with **1–3 second** response latency.
- Implemented **SHA-256 response caching**, a **20 requests/minute** rate limit, and a Redis-backed rate limiter with in-memory failover; deployed the platform and execution runner on Render.

Skin Disease Detection System Using CNN [GitHub](#)

Tech Stack: Python, TensorFlow, EfficientNetB3, OpenCV, Scikit-Learn, CNN

- Created a skin lesion classification pipeline on the HAM10000 dataset containing **10,015** dermoscopic images and restructured labels into **3** classes: benign, melanoma, and non-melanoma.
- Prevented data leakage through patient-id based train-validation-test splitting and used a dual-stream preprocessing pipeline using **CLAHE** enhancement and **black-hat morphology**-based hair removal.
- Fine-tuned **EfficientNetB3** using transfer learning, Focal Loss, Adam optimization, and **Test-Time Augmentation**.
- Achieved **90%** test accuracy with F1-scores of **0.89** (benign), **0.86** (melanoma), and **0.94** (non-melanoma), and evaluated performance using confusion matrices and per-class precision/recall analysis.

EDUCATION

Master of Computer Applications Sikkim Manipal Institute of Technology	2024 – Present CGPA: 8.16
Bachelor of Computer Applications Techno India University	2021 – 2024 CGPA: 8.39
Higher Secondary (Class XII) Maynaguri High School	2021 Percentage: 88.6%

CERTIFICATIONS

Android Developer Virtual Internship (Google & AICTE EduSkills) Grade: Excellent	Apr 2024 – Jun 2024 Credential
IBM Professional Certificate: Full Stack Software Developer	Credential
IBM Professional Certificate: Data Science and AI	Credential